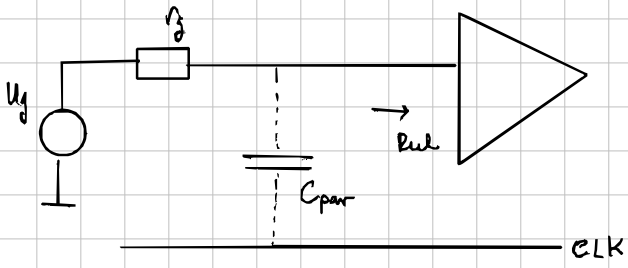
DIFERENCIJSKA POJAČALA

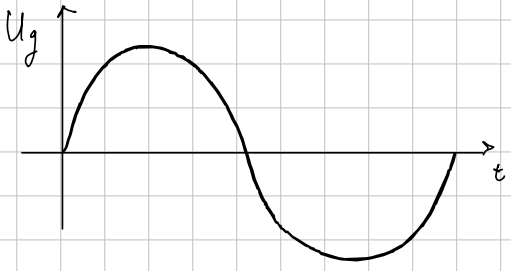
- ulazni stupanj u OP, komparatore i stabilizatorima

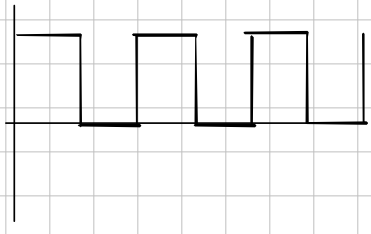
⇒ integrirana tehnika

• 2 ulaza → 1 ili 2 izlaza → /⇉

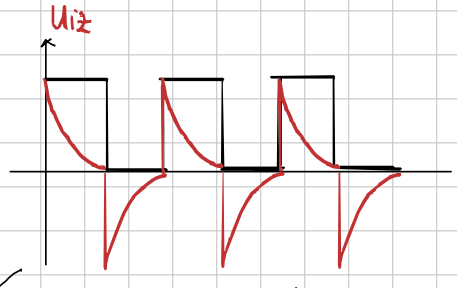
Primjer:



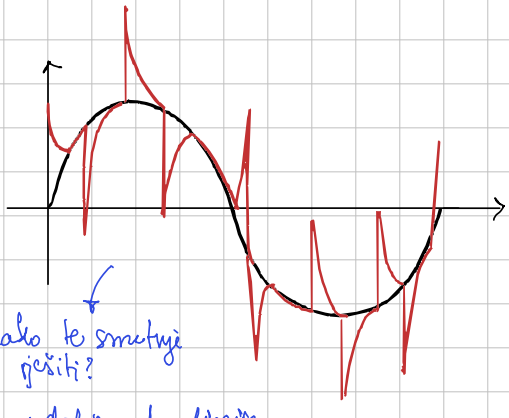




⇒



↳ taj signal se preslikava na ulaz OP (izlaz CR mreže je ulaz OP)

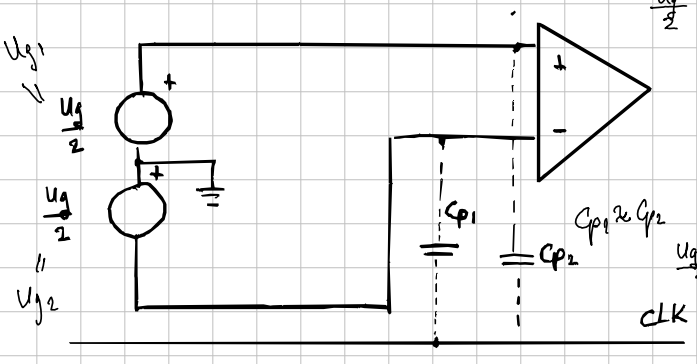


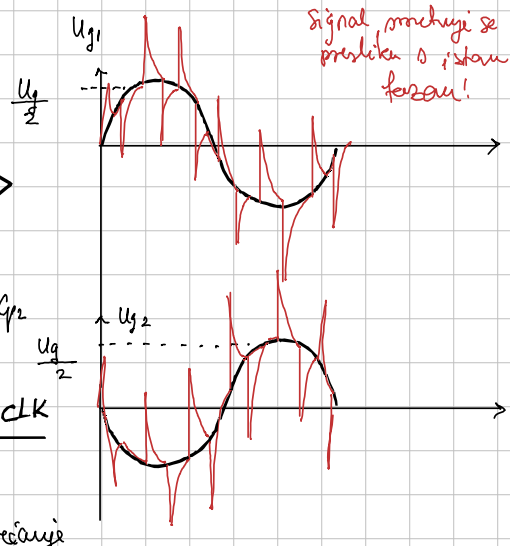
⇒ izlaz OP-a je ulazni sinusni napon superponiran izlaznim naponom RC mreže

kako te smetnje ispeći?

- udaljenost linija
- oklapanje linija
- sklopovski smanjimo smetnje

⇒ DIFERENCIJSKO POJAČALO

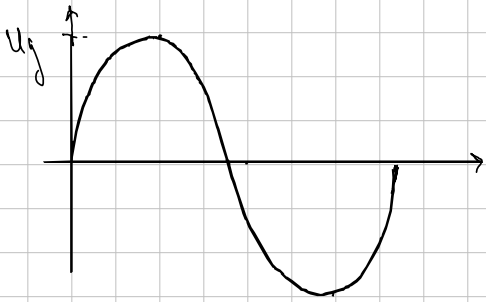




signal smetnji se preslika s istom fazom!

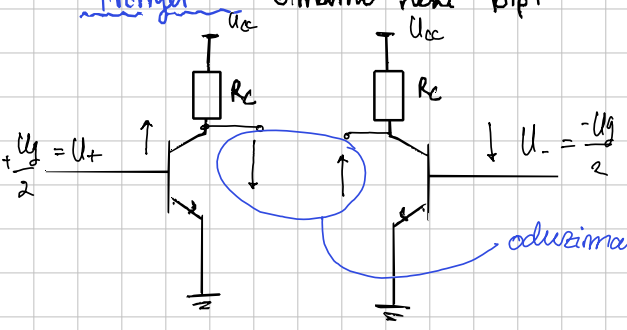
$U_{g1} - U_{g2} = \frac{U_g}{2} - \left(-\frac{U_g}{2}\right) = U_g$ dvostruko povećanje amplitude

, ali šiljci se ponište jer su istofazni signali!



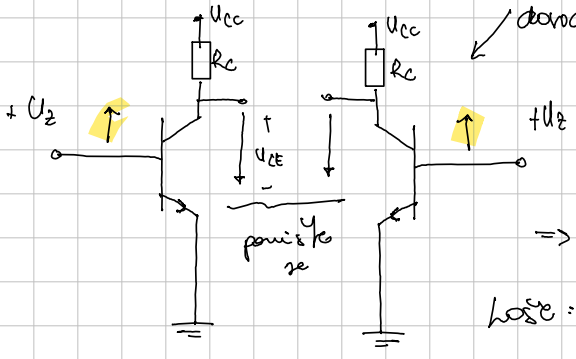
tako radi diferencijsko pojačalo

Primjer: Imamo neki bipT



oduzimanjem dobivamo dvostruko veći signal

DIF. SIGNAL NA ULAZU



dandimo smetnje = izlazni signal

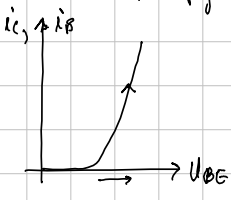
zbog jednakosti $C_{p1} = C_{p2}$ → ista faza (i amplituda)

⇒ doprinos na izlazu je ∅

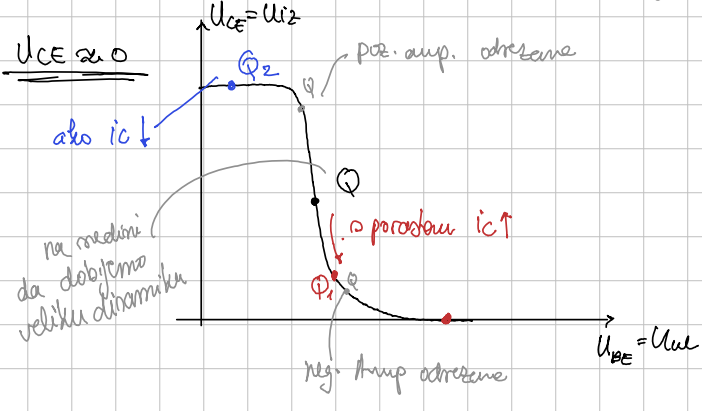
Loše: smetnja može imati sporu promjenu (npr 50 Hz), a mi želimo izmjeriti napone na 200 Hz

$U_{CE} = U_{CC} - i_C \cdot R_C$ ⇒ $U_{CE} \approx 0$

Karakt PN spoja:



$i_C \uparrow$ exp



⇒ zaključak: ovako pojačalo je beskorisno = ne možemo ga koristiti kao diferencijalno

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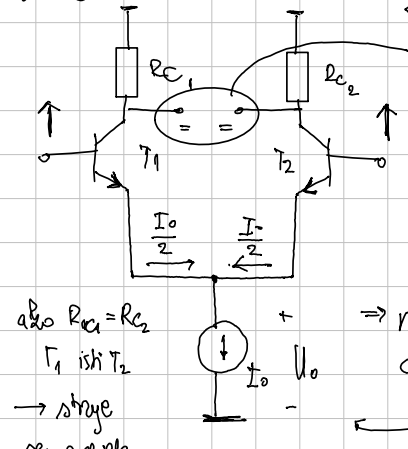
RJEŠENJE:



struju drugačije podešavamo: strujni izvor

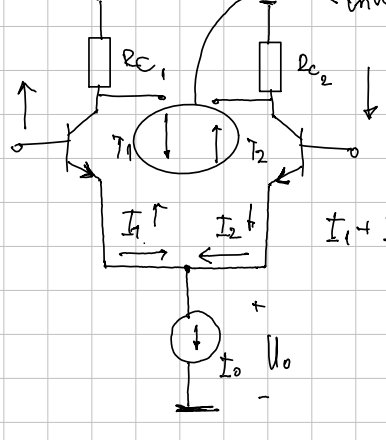
dobivamo konstantan napon ⇒ povećanje ≈ 0

ako $R_{C_1} = R_{C_2}$, T_1 isti T_2 → struje se podjele 1:1

⇒ napon na emiteru se podesi tako da zadovolje ono što diktira izvor

U_{CE} = konst. → jer je $I_C = \frac{I_0}{2}$

Što ako imamo protufazne promjene:



$I_1 + I_2 = I_0$